

Weak Sequence Dependence via Mallows-Penalized Bag-of-Words Models

Abstract

We propose a minimal extension of the bag-of-words energy model for word co-occurrence that introduces weak sequential dependence through a Mallows-type penalty on the symmetric group. The observed word ordering is penalized by its Kendall τ distance from a reference (normal) ordering, with coupling strength θ . At $\theta = 0$ the model reduces to the permutation-invariant bag-of-words theory; at small θ a perturbative expansion yields a directional correction to the pointwise mutual information (PMI) that captures word-order preference. The Mallows partition function is a q -factorial, and the pairwise structure of the Kendall distance meshes naturally with pairwise co-occurrence interactions, keeping the model analytically tractable.

1 The bag-of-words baseline

Consider a vocabulary of N tokens and a text window of length L with word sequence $(w_1, \dots, w_L)$. The bag-of-words energy model assigns

$$p_{\text{BoW}}(w_1, \dots, w_L) \propto \exp\left(-\sum_{i,j} G_{ij} n_i n_j\right), \quad (1)$$

where $n_i = \sum_{p=1}^L \delta_{i,y_p} - P_i$ is the count fluctuation of token i and G_{ij} is a coupling matrix. This distribution is invariant under permutations of the positions $\{1, \dots, L\}$: it depends only on the multiset of tokens, not their ordering.

The central result of the bag-of-words theory is that embedding vectors $\mathbf{v}_i$ fitted to this model satisfy $\mathbf{v}_i \cdot \mathbf{v}_j \propto \text{PMI}(i, j)$, with the dot product encoding the pointwise mutual information.

2 Breaking permutation symmetry: the Mallows penalty

2.1 Setup

A word sequence $(w_1, \dots, w_L)$ of distinct tokens determines an ordering $\sigma \in S_L$. We introduce a *normal ordering* $\sigma_0 \in S_L$ —a reference permutation representing the “expected” order of the tokens—and penalize deviations from it. The extended model is

$$p(w_1, \dots, w_L; \sigma) \propto \exp\left(-\sum_{i,j} G_{ij} n_i n_j - \theta d(\sigma, \sigma_0)\right) \quad (2)$$

where $d(\sigma, \sigma_0)$ is a distance on the symmetric group S_L and $\theta \geq 0$ controls the strength of the ordering preference. At $\theta = 0$ we recover (1) exactly.

This is a *Mallows model* [1] coupled to an energy-based co-occurrence model. The Mallows model assigns $p(\sigma \mid \sigma_0, \theta) \propto e^{-\theta d(\sigma, \sigma_0)}$ and is a standard exponential family on S_L .

2.2 Choice of distance: Kendall τ

Three natural right-invariant metrics on S_L are:

Distance	Definition	Structure
Kendall τ	# pairwise inversions	pairwise: $\sum_{i < j} \mathbb{I}[\sigma \text{ inverts } (i, j)]$
Cayley	min # transpositions	$L - \#\text{cycles}(\sigma \sigma_0^{-1})$
Hamming	# displaced positions	$\sum_i \mathbb{I}[\sigma(i) \neq \sigma_0(i)]$

We choose the **Kendall τ distance** d_K for two reasons:

1. **Pairwise decomposition.** The Kendall distance is a sum over pairs, so the ordering penalty has the same two-body structure as the co-occurrence energy $\sum G_{ij} n_i n_j$. The full model remains a sum of pairwise terms—one set over token types, one set over positions.
2. **Closed-form partition function.** The Mallows partition function with Kendall distance factorizes as a q -factorial:

$$Z_{\text{Mallows}}(\theta) = \sum_{\sigma \in S_L} e^{-\theta d_K(\sigma, \sigma_0)} = \prod_{k=1}^L [k]_q = [L]_q!, \quad (3)$$

where $q = e^{-\theta}$ and $[k]_q = \frac{1-q^k}{1-q}$ is the q -analog of k . This closed form makes the model analytically tractable.

3 Perturbative expansion at small θ

3.1 Free energy correction

Write the full partition function as

$$Z(\theta) = \sum_{\{w\}} \sum_{\sigma \in S_L} \exp\left(-\sum_{i,j} G_{ij} n_i n_j - \theta d_K(\sigma, \sigma_0)\right). \quad (4)$$

At $\theta = 0$, the sum over σ gives $L!$ (all orderings equally weighted), recovering the bag-of-words model up to normalization. For small θ , the free energy $F(\theta) = -\log Z(\theta)$ admits the expansion

$$F(\theta) = F_{\text{BoW}} + \theta \langle d_K(\sigma, \sigma_0) \rangle_{\text{BoW}} + \frac{\theta^2}{2} \left(\langle d_K^2 \rangle_{\text{BoW}} - \langle d_K \rangle_{\text{BoW}}^2 \right) + O(\theta^3), \quad (5)$$

where $\langle \cdot \rangle_{\text{BoW}}$ denotes expectation under the $\theta = 0$ (bag-of-words) distribution. The first-order correction is the expected number of inversions relative to σ_0 in the unordered model.

3.2 Directional PMI

The ordering penalty introduces an asymmetry between the pair (w_i, w_j) appearing as $(\dots w_i \dots w_j \dots)$ versus $(\dots w_j \dots w_i \dots)$. To first order in θ , the effective PMI splits into symmetric and antisymmetric parts:

$$\text{PMI}_{\text{eff}}(i, j) = \underbrace{\text{PMI}_{\text{BoW}}(i, j)}_{\text{symmetric}} + \theta \cdot \underbrace{\text{PMI}_{\rightarrow}(i, j)}_{\text{directional}} + O(\theta^2), \quad (6)$$

where $\text{PMI}_{\text{BoW}}(i, j)$ is the standard (permutation-invariant) pointwise mutual information and

$$\text{PMI}_{\rightarrow}(i, j) = \log \frac{p(i \text{ precedes } j \mid \text{co-occur})}{p(i \text{ follows } j \mid \text{co-occur})} \quad (7)$$

measures the tendency of token i to appear before token j , conditioned on their co-occurrence. Note that $\text{PMI}_{\rightarrow}(i, j) = -\text{PMI}_{\rightarrow}(j, i)$: the directional correction is antisymmetric.

3.3 Implications for embedding geometry

In the bag-of-words theory, embedding dot products encode the symmetric PMI: $\mathbf{v}_i \cdot \mathbf{v}_j \propto \text{PMI}_{\text{BoW}}(i, j)$. The directional correction (6) suggests that, at $\theta > 0$, the embedding must encode both symmetric and antisymmetric information. A natural ansatz is to decompose each embedding into a symmetric part $\mathbf{v}_i^{(s)}$ and a directional part $\mathbf{v}_i^{(d)}$:

$$\mathbf{v}_i^{(s)} \cdot \mathbf{v}_j^{(s)} \propto \text{PMI}_{\text{BoW}}(i, j), \quad \mathbf{v}_i^{(d)} \cdot \mathbf{v}_j^{(s)} - \mathbf{v}_j^{(d)} \cdot \mathbf{v}_i^{(s)} \propto \theta \text{PMI}_{\rightarrow}(i, j). \quad (8)$$

This is analogous to decomposing a tensor into symmetric and antisymmetric parts, or to the word/context vector distinction in skip-gram models—but here the decomposition is derived from a symmetry-breaking principle rather than imposed by algorithmic design.

4 The normal ordering σ_0

The choice of reference ordering σ_0 is a modeling decision. Three natural options within the existing framework:

Discourse-vector projection. In the RAND-WALK model [2], the discourse vector $\mathbf{c}_t$ provides a salience ordering at each time step: rank words by $\langle \mathbf{c}_t, \mathbf{v}_{w_i} \rangle$. The most discourse-relevant word comes first. This makes σ_0 context-dependent.

Frequency ordering. Rank tokens by unigram probability P_i (most frequent first). This gives a universal, context-independent reference connected to Zipf’s law. The Kendall distance from frequency order measures how “surprising” the word arrangement is.

Family of normal orders. Let σ_0 be drawn from a prior distribution—for instance, a Mallows distribution centered on a syntactic template (SVO order). Marginalizing over σ_0 :

$$p(w_1, \dots, w_L) = \sum_{\sigma_0} p(\sigma_0) \frac{1}{Z(\theta) Z_{\text{BoW}}} \exp\left(-\sum_{i,j} G_{ij} n_i n_j - \theta d_K(\sigma, \sigma_0)\right). \quad (9)$$

Mallows mixtures are well-studied and admit efficient inference [3], so this remains tractable.

5 Discussion

The proposed model has several attractive features:

Controlled symmetry breaking. The parameter θ interpolates continuously between the fully permutation-invariant bag-of-words model ($\theta = 0$) and a strongly order-dependent model ($\theta \rightarrow \infty$). The perturbative expansion in θ gives systematic corrections to the PMI, with the zeroth-order term recovering the established theory exactly.

q -deformation. The Mallows partition function (3) is a q -deformation of the factorial, with $q = e^{-\theta}$. At $\theta = 0$, $q = 1$ and $[L]_q! = L!$ (full permutation symmetry). As θ increases, $q < 1$ and the effective number of accessible orderings shrinks. This is the combinatorial analogue of lowering the temperature in a spin system.

Pairwise consistency. The Kendall distance decomposes into independent pair contributions, preserving the pairwise structure of the co-occurrence energy. The full model is a product of two-body interactions—one in “token space” and one in “position space.”

Open questions. (1) What is the exact form of the corrected embedding geometry at finite θ —does the antisymmetric part of the PMI induce a symplectic-like structure on the embedding space? (2) Can the q -factorial structure be connected to quantum group symmetries of the embedding? (3) Does the perturbative expansion converge, or is there a phase transition at some critical θ_c where the permutation symmetry breaks spontaneously?

References

- [1] C. L. Mallows, “Non-null ranking models. I,” *Biometrika*, vol. 44, no. 1/2, pp. 114–130, 1957.
- [2] S. Arora, Y. Li, Y. Liang, T. Ma, and A. Risteski, “A Latent Variable Model Approach to PMI-based Word Embeddings,” *Transactions of the Association for Computational Linguistics*, vol. 4, pp. 385–399, 2016. arXiv:1502.03520.
- [3] T. Lu and C. Boutilier, “Learning Mallows models with pairwise preferences,” *Proceedings of the 28th International Conference on Machine Learning*, pp. 145–152, 2011.